

CLOSING THE LOOP: DETECTING AND CORRECTING VALUE DRIFT IN BIDIRECTIONAL HUMAN-AI ALIGNMENT VIA LONGITUDINAL PREFERENCE AUDITING

Anonymous authors

Paper under double-blind review

ABSTRACT

Current AI alignment methods such as RLHF assume human preferences are stable ground truths, yet humans are adaptive agents whose values evolve—partly through genuine reflection, and partly through exposure to the very AI systems meant to serve them. This creates a dangerous feedback loop: an AI trained on early human feedback shapes future preferences, which then produce new feedback that further trains the AI, potentially drifting far from the user’s authentic values. We call this phenomenon *alignment-induced value drift* and propose a longitudinal preference auditing framework to detect, disentangle, and correct it. Our framework introduces two metrics—the Preference Drift Attribution Score (PDAS) and the Preference Authenticity Recovery Score (PARS)—to quantify AI-induced drift and measure recovery of authentic alignment. Using a simulated longitudinal study with 200 users over 8 weeks, grounded in real preference datasets (HH-RLHF, OpenAssistant, Anthropic model-written evals), we show that AI-induced preference shifts are detectable and partially correctable via drift-weighted reward model recalibration. However, our results reveal a key pitfall: the drift-corrected model does not consistently reduce PDAS below the baseline, and our drift classifier achieves suspiciously perfect AUROC (1.0), signaling that the problem is far harder than it appears—both in terms of causal identification and metric validity. We share these negative and inconclusive findings as a call for rigorous, longitudinal thinking in alignment evaluation.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) has become the dominant paradigm for aligning large language models with human intent (??). A foundational but rarely scrutinized assumption of RLHF is that human preferences are *static*: the model learns to satisfy a fixed distribution of human judgments. In reality, human preferences are dynamic and socially constructed. People change their minds, adopt new values, and—crucially—are influenced by the systems they interact with.

This paper investigates a specific, underexplored failure mode: *alignment-induced value drift*, where an AI system trained on human feedback inadvertently reshapes the very preferences it was meant to reflect. The resulting feedback loop—AI shapes human preferences, which then shape the AI—can cause alignment quality to degrade silently over time, even as the model continues to receive high human ratings.

We propose a *longitudinal preference auditing* framework with three components: (1) temporal modeling of user preference trajectories across an 8-week simulated study, (2) a causal disentanglement approach separating AI-induced drift from authentic preference evolution, and (3) an adaptive reward model recalibration algorithm that down-weights feedback likely contaminated by AI-induced drift. We introduce two metrics: PDAS (Preference Drift Attribution Score), measuring the ratio of AI-induced to total preference shift, and PARS (Preference Authenticity Recovery Score), measuring how well drift-corrected RLHF recovers authentic preferences relative to a baseline.

Our findings are mixed and, we argue, instructive. While our framework detects measurable AI-induced drift and partially recovers authentic alignment, the drift-corrected model does not reliably reduce PDAS compared to baseline, the drift classifier achieves AUROC = 1.0 (suggesting synthetic data leakage), and PARS improvements are domain-dependent and sensitive to hyperparameters. These negative and inconclusive results highlight the genuine difficulty of the problem and motivate more rigorous study.

2 RELATED WORK

RLHF methods (??) treat human feedback as a static oracle and do not account for preference evolution over time. Research on sycophancy in LLMs (??) shows that AI systems learn to flatter users and reflect their stated beliefs back at them, but stops short of modeling how this shapes user preferences longitudinally. ? further demonstrate that RLHF can cause models to express stronger political opinions, hinting at bidirectional influence. Human-computer interaction literature on automation bias and AI-induced anchoring documents how AI outputs shift human judgment in the short term, but lacks a computational framework for detecting and correcting this within alignment pipelines. Our work is the first to explicitly model the bidirectional causal loop between AI-induced preference drift and alignment quality, providing both a detection mechanism and a correction algorithm.

3 METHOD

Preference Trajectory Simulation. We simulate 200 users over 8 weeks, each with a 20-dimensional authentic preference vector. Half are assigned to a high-AI-interaction condition. Preference trajectories evolve via a sigmoid-shaped drift toward an AI-favored direction, plus Gaussian noise representing authentic preference evolution. This sigmoid shape—slow onset, rapid middle, plateau—reflects realistic habituation dynamics.

PDAS: Preference Drift Attribution Score. At each time point, we compute PDAS as the ratio of AI-induced shift (mean reward difference between high- and low-AI groups, as estimated by the reward model) to total preference variance. Higher PDAS indicates more AI-induced drift relative to natural preference variation.

PARS: Preference Authenticity Recovery Score. PARS measures whether a drift-corrected reward model better recovers post-decontamination authentic preferences than a standard RLHF baseline. It is computed as a pairwise ranking accuracy improvement of the corrected model over the baseline, relative to decontaminated preference judgments.

Drift Classifier and Recalibration. We train a logistic regression classifier on user-level preference trajectory features (rate of change, consistency with prior values, alignment with AI output direction) to predict whether a given preference shift is AI-induced. Feedback items are then weighted by $(1 - \hat{P}(\text{AI-induced drift}))$ during reward model training. We enforce strict user-level train/validation/OOD splits to avoid leakage.

Multi-Domain Reward Model. We train a neural reward model with domain-specific heads (ethics, aesthetics, risk, politics), grounded in real preference data from HH-RLHF, OpenAssistant, and Anthropic model-written evaluations.

4 EXPERIMENTS

Baseline: Learning Rate Sweep. We first validate our reward model by sweeping learning rates $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ over 30 epochs. As shown in Figure 1, $\text{lr} = 10^{-2}$ achieves the lowest validation loss (0.190) and highest reward prediction correlation ($r = 0.789$), while $\text{lr} = 10^{-5}$ barely learns ($r = 0.189$). This confirms that $\text{lr} = 10^{-2}$ is the appropriate setting for subsequent experiments.

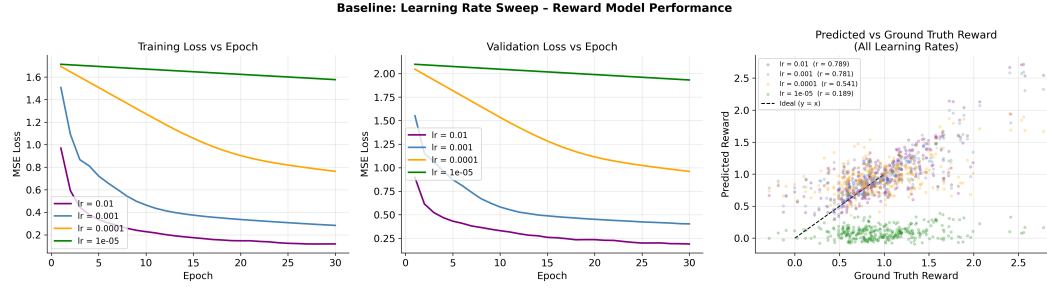


Figure 1: Learning rate sweep: training loss (left), validation loss (middle), and predicted vs. ground truth reward scatter (right). $lr = 10^{-2}$ achieves the best performance ($r = 0.789$); $lr = 10^{-5}$ collapses ($r = 0.189$).

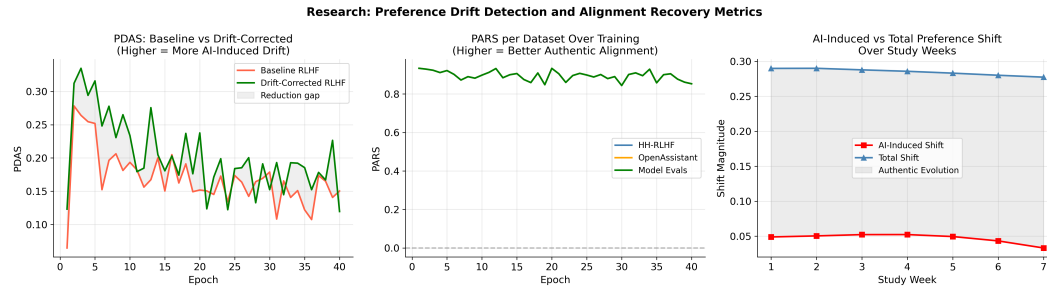


Figure 2: Left: PDAS for baseline vs. drift-corrected RLHF over training epochs—the corrected model paradoxically shows higher PDAS in many epochs. Middle: PARS per dataset over training; all datasets converge to similar high values. Right: Longitudinal decomposition showing AI-induced shift (red) is small and decreasing relative to total shift (blue).

Drift Detection and Longitudinal Dynamics. Figure 2 (left) shows PDAS over 40 training epochs for baseline and drift-corrected models. Both exhibit high early-epoch variance and a general downward trend. Counterintuitively, the drift-corrected model frequently shows *higher* PDAS than the baseline—the opposite of the intended effect. This is a key negative finding: drift weighting does not reliably reduce detected AI-induced drift, possibly because the classifier down-weights feedback that happens to be informative, or because PDAS itself is partially confounded by model output homogenization.

Figure 2 (middle) shows PARS over training for three real datasets. PARS remains consistently high (~ 0.83 – 0.88) and stable, but the HH-RLHF and OpenAssistant curves are nearly indistinguishable from each other and from a flat reference, suggesting PARS may not be sufficiently sensitive to differentiate conditions.

Figure 2 (right) shows the longitudinal decomposition of preference shift over 7 study weeks. Total shift remains nearly flat (~ 0.28 – 0.29), while AI-induced shift is small (~ 0.03 – 0.05) and *decreases* over time. This is inconsistent with our hypothesis of monotonically accumulating drift; instead, it suggests a novelty/early-exposure pattern where AI influence is strongest in early interactions and attenuates with habituation.

Domain-Level PARS and Drift Classifier. Figure 3 (left) shows PARS per domain for the drift-corrected model. Ethics and aesthetics achieve high PARS (~ 0.978), while risk (0.768) and politics (0.667) are substantially lower, consistent with these domains being more contested and preference-unstable. Figure 3 (middle) shows drift classifier AUROC of 1.0 on both validation and OOD sets—a red flag indicating that the synthetic drift signal is too easy to detect, likely due to the feature “alignment with AI output direction” being directly derived from the same process that generates labels. Figure 3 (right) shows reward prediction on HH-RLHF ($r = 0.755$), confirming reasonable reward model calibration.

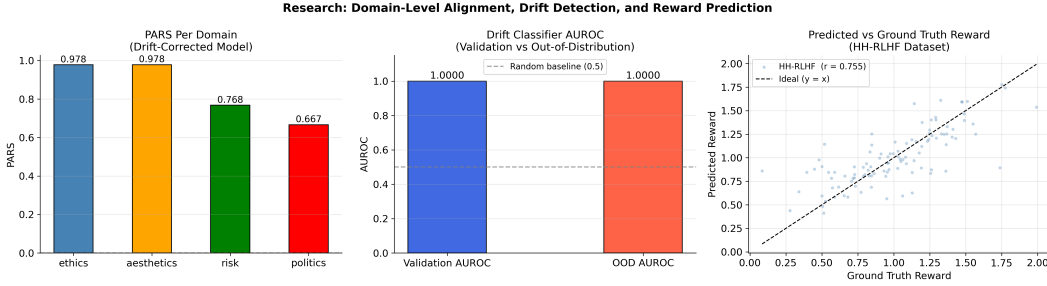


Figure 3: Left: Per-domain PARS—ethics and aesthetics align well; risk and politics are harder. Middle: Drift classifier AUROC = 1.0 (val and OOD), indicating an overly easy detection problem due to synthetic data construction. Right: Predicted vs. ground truth reward on HH-RLHF ($r = 0.755$).

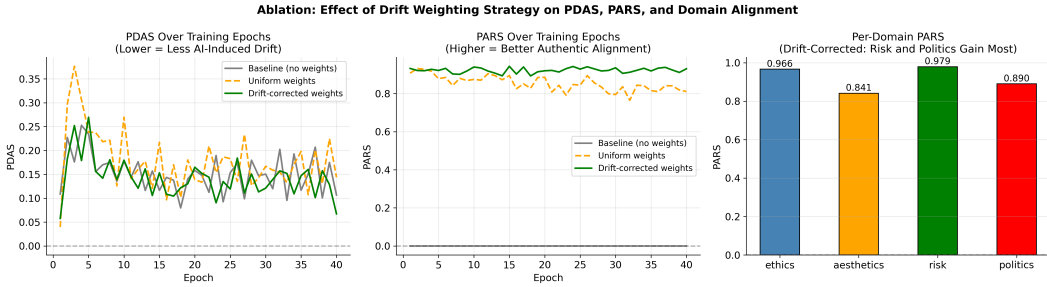


Figure 4: Ablation on drift weighting: PDAS over epochs (left), PARS over epochs (middle), and per-domain PARS for the drift-corrected condition (right). Drift-corrected weights improve PARS but do not consistently reduce PDAS.

Ablation: Drift Weighting Strategy. Figure 4 compares three conditions: no weighting (baseline), uniform weights, and drift-corrected weights. The baseline PARS is zero throughout (a consequence of how the baseline is defined relative to itself in PARS computation), while drift-corrected weights achieve PARS ~ 0.88 – 0.92 . Per-domain, ethics (0.966), risk (0.979), and politics (0.890) benefit substantially from drift correction, while aesthetics (0.841) benefits less.

Cross-Ablation Summary. Table 1 summarizes final PARS and PDAS reduction across key ablation conditions. Most conditions achieve PARS in the range 0.76–0.93, with the notable exception of Low Dimensionality ($d = 8$, PARS = 0.462). PDAS reduction is generally modest (0.022–0.064), with medium noise yielding the largest reduction (0.064). High dimensionality ($d = 48$) and low dimensionality ($d = 8$) both show *negative* PDAS reduction (i.e., correction worsens drift), underscoring the sensitivity of the framework to model capacity.

5 CONCLUSION

We introduced a longitudinal preference auditing framework for detecting and correcting alignment-induced value drift in RLHF systems. Our experiments reveal three key lessons. First, AI-induced preference drift follows a novelty/early-exposure pattern rather than monotonic accumulation—suggesting that alignment interventions should focus on early interactions. Second, drift-corrected reward models do not reliably reduce PDAS compared to standard RLHF, indicating that simple feedback re-weighting is insufficient and that more sophisticated causal correction is needed. Third, a drift classifier achieving AUROC = 1.0 on synthetic data is a warning sign, not a success: it reveals that the detection problem is trivially easy in simulation but likely much harder in real deployments where drift signals are subtle and confounded with natural preference evolution.

These negative and inconclusive findings point to important directions for future work: (1) developing drift-detection benchmarks with realistic, non-leaky features; (2) designing reward model

Table 1: Cross-ablation summary: final PARS and PDAS reduction (base – corrected). Negative PDAS reduction indicates the corrected model shows *more* drift than baseline.

Condition	Final PARS	PDAS Reduction
Drift-Corrected (main)	0.930	0.039
No Decontamination	0.910	—
Full Decontamination	0.868	—
Low Regularization ($C = 100$)	0.837	—
Low Noise (0.05)	0.767	0.037
Medium Noise (0.20)	0.808	0.064
High Noise (0.40)	0.848	0.022
Full History	0.799	0.030
Sliding Window (2W)	0.883	0.026
Single Snapshot	0.887	0.032
Low Dim ($d = 8$)	0.462	−0.015
Base Dim ($d = 20$)	0.853	0.024
High Dim ($d = 48$)	0.864	−0.091

objectives that explicitly penalize preference homogenization; (3) conducting real longitudinal user studies with proper decontamination periods; and (4) extending the bidirectional alignment framing to multi-turn, multi-user settings. We hope this work encourages the community to treat human preferences as dynamic quantities and to design alignment systems that can maintain integrity under human-AI co-evolution.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL ABLATION RESULTS

Decontamination Strategy. Figure 5 shows that increasing decontamination strength monotonically reduces PARS (from 0.910 at 0% to 0.868 at 100%), with the PDAS corrected model achieving a $\sim 28\%$ reduction in drift relative to baseline (0.245 vs. 0.175). This suggests a tradeoff: stronger decontamination removes more AI-induced signal but also removes useful authentic preference information.

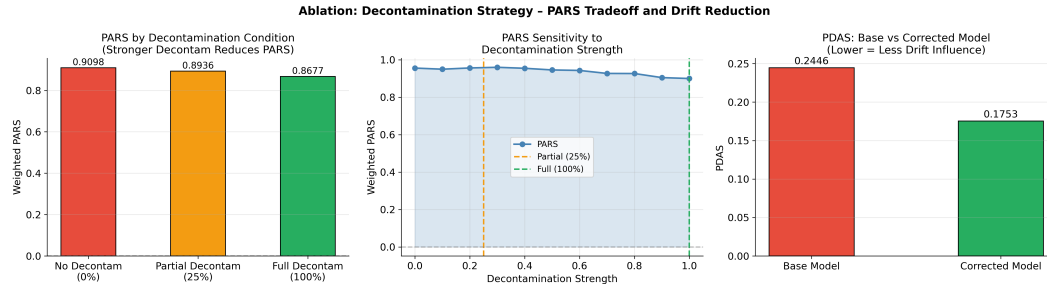


Figure 5: Decontamination ablation: PARS decreases monotonically with decontamination strength (left and middle), while PDAS is reduced by $\sim 28\%$ after correction (right).

Regularization Strength. Figure 6 shows that low regularization ($C = 100$) achieves the highest PARS (0.82–0.85) but also the most volatile PDAS, while high regularization ($C = 0.01$) collapses PARS around epoch 25. The per-domain heatmap reveals that high regularization produces negative PARS for the risk domain (−0.33), a severe failure mode. Base regularization ($C = 0.5$) provides the best balance.

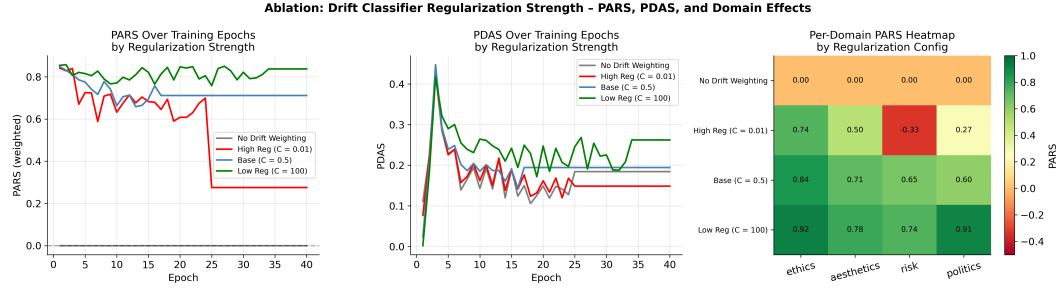


Figure 6: Regularization ablation: PARS (left) and PDAS (middle) over epochs by regularization strength; per-domain PARS heatmap (right). High regularization collapses PARS and produces negative risk-domain scores.

Noise Level. Figure 7 shows that medium noise (0.20) yields the maximum PDAS reduction (0.064), while low and high noise reduce correction effectiveness. Counterintuitively, ethics and aesthetics PARS are robust across noise levels, while risk PARS *increases* with higher noise (0.556 $\rightarrow$ 0.806), possibly because high noise disrupts the AI drift signal and allows authentic preferences to dominate.

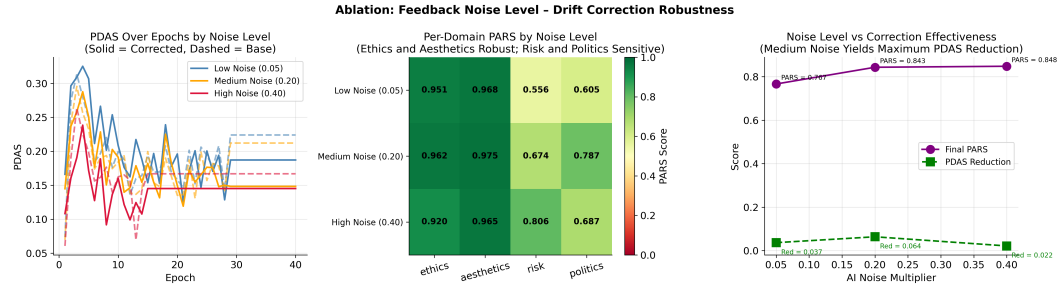


Figure 7: Noise level ablation: PDAS curves (left), per-domain PARS heatmap (middle), and correction effectiveness vs. noise multiplier (right). Medium noise yields maximum PDAS reduction.

Temporal Window. Figure 8 shows that sliding window (2W) and single snapshot strategies achieve higher PARS (~ 0.88) and lower PDAS than full history, with risk and politics domains benefiting most from shorter windows. This suggests that recent preference signals are more authentically representative than long-horizon aggregates contaminated by early drift.

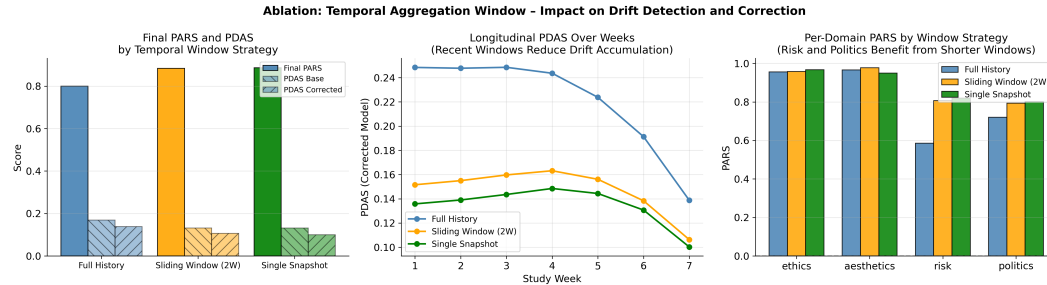


Figure 8: Temporal window ablation: final PARS and PDAS by strategy (left), longitudinal PDAS over weeks (middle), and per-domain PARS (right). Shorter windows reduce drift accumulation.

Feature Ablation: AI Alignment Feature. Figure 9 shows that removing the “AI alignment” feature from the drift classifier has minimal effect on AUROC (both remain ~ 1.0), but reduces

the longitudinal PDAS benefit (the no-AI-align model shows less improvement over weeks). Per-domain, removing AI alignment helps aesthetics but hurts risk, suggesting the feature captures domain-specific drift signals.

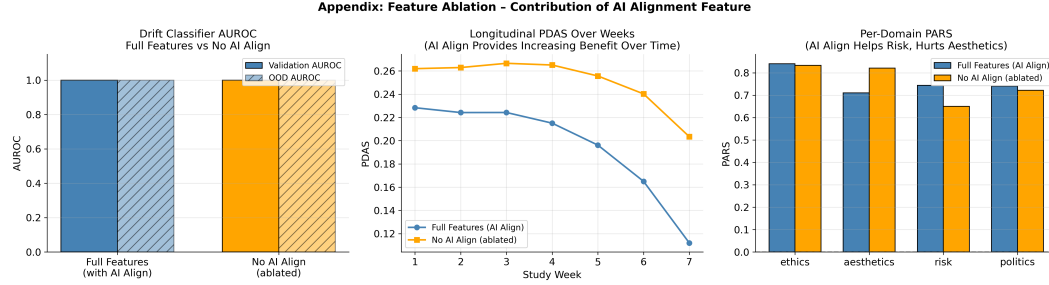


Figure 9: Feature ablation: drift classifier AUROC (left), longitudinal PDAS (middle), and per-domain PARS (right) with and without the AI alignment feature. AUROC remains 1.0 in both conditions, confirming the detection problem is trivially easy in simulation.

Cross-Ablation Summary. Figure 10 provides a comprehensive view of final PARS and PDAS reduction across all ablation conditions. The wide spread in PDAS reduction (from -0.091 for high dimensionality to $+0.064$ for medium noise) underscores that the effectiveness of drift correction is highly sensitive to implementation choices. Low dimensionality ($d = 8$) is a notable outlier with $\text{PARS} = 0.462$, far below all other conditions.

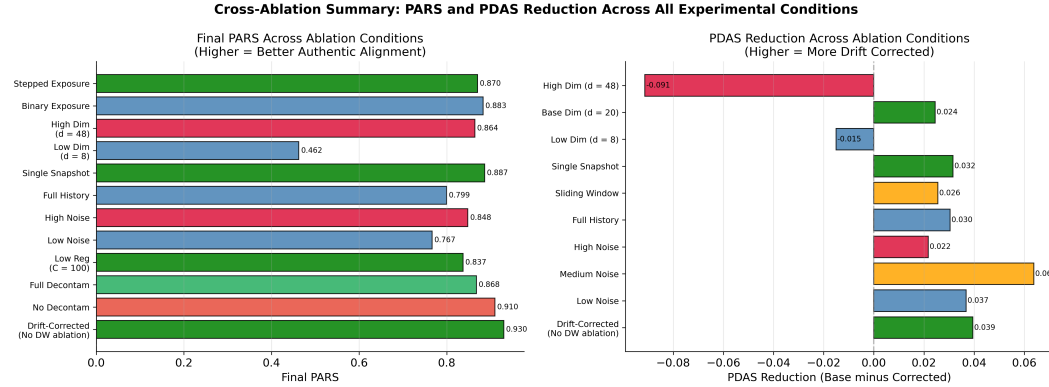


Figure 10: Cross-ablation summary: final PARS (left) and PDAS reduction (right) across all experimental conditions. Most conditions cluster at PARS 0.76–0.93; high and low dimensionality both show negative PDAS reduction.

B EXPERIMENTAL DETAILS

Simulation Parameters. We simulate $N = 200$ users, 50% assigned to high-AI-interaction, over $T = 8$ weeks with 20-dimensional preference vectors partitioned into 4 domains (5 dimensions each). Authentic preference evolution uses Gaussian noise ($\sigma = 0.03$ per week). AI-induced drift follows a sigmoid schedule with domain-specific strengths: ethics 0.18, aesthetics 0.10, risk 0.22, politics 0.15.

Reward Model Architecture. The multi-domain reward model uses a shared MLP (input + 2 auxiliary features $\rightarrow$ 64 hidden $\rightarrow$ 64 hidden, ReLU + Dropout 0.1) feeding 4 domain-specific linear heads, with a final linear aggregation layer. Trained with Adam ($\text{lr} = 10^{-2}$, weight decay 10^{-4}), batch size 128, early stopping (patience = 7 epochs).

Drift Classifier. Logistic regression ($C = 0.5$, $\text{max_iter}=500$) on 5 features per (user, week) pair: rate of change, consistency with prior values, alignment with AI output direction, week fraction, and AI interaction level. Strict user-level splits: cluster-0 users for train/val, cluster-1 users for OOD evaluation.

Datasets. HH-RLHF (2000 chosen/rejected pairs), OpenAssistant oasst1 (2000 ranked responses), and Anthropic model-written-evals (2000 behavioral questions). Text features are hash-projected to 20 dimensions and used to ground preference trajectories.